

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 is only forming an OSPF adjacency with R1 because only S0/0/0 on R2 is active for OSPF.

S0/0/1 is still passive, so R3 does not become an OSPF neighbor with R2.

Therefore, R3 reaches 192.168.2.0/24 indirectly through R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 R2(config)# router ospf 1

 R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To allow OSPF to assign more accurate costs to high-speed links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24: R1 serial cost 781 + R3 G0/0 cost 1 = 782

For 192.168.23.0/30: The route crosses two serial links, so the total cost is $781 + 64 = 845$

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, Because after setting all serial interfaces to bandwidth 128, each serial link has a cost of about 781. So the total path cost is $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the ip ospf cost 1565 command increased the direct path cost on R1's S0/0/1 interface.

That made the route through R2 lower in total OSPF cost, so OSPF selected that path.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

It is important to control the router ID because it uniquely identifies each OSPF router and helps provide stable neighbor relationships and easier troubleshooting.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election process is not a concern in this lab because the OSPF links are mainly point-to-point serial links, where DR/BDR elections do not occur.

3. Why would you want to set an OSPF interface to passive?

You would set an OSPF interface to passive to prevent hello packets and neighbor formation on interfaces without OSPF neighbors while still advertising the connected network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas makes OSPF hierarchical and allows the creation of scalable networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

The command is `network 10.0.0.0 0.255.255.255 area 0`.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used because it has the lowest administrative distance.